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# Why Language Models Hallucinate

Adam Tauman Kalai* OpenAI
Ofir Nachum OpenAI
Santosh S. Vempala† Georgia Tech
Edwin Zhang OpenAI

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**Abstract**

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such "hallucinations" persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty, and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This "epidemic" of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

## 1 Introduction

Language models are known to produce overconfident, plausible falsehoods, which diminish their utility and trustworthiness. This error mode is known as "hallucination," though it differs fundamentally from the human perceptual experience. Despite significant progress, hallucinations continue to plague the field, and are still present in the latest models (OpenAI, 2025a). Consider the prompt:

What is Adam Tauman Kalai's birthday? If you know, just respond with DD-MM.

On three separate attempts, a state-of-the-art open-source language model[1] output three incorrect dates: "03-07", "15-06", and "01-01", even though a response was requested only if known. The correct date is in Autumn. Table 1 provides an example of more elaborate hallucinations.

Hallucinations are an important special case of *errors* produced by language models, which we analyze more generally using computational learning theory (e.g., Kearns and Vazirani, 1994). We consider general sets of *errors* $\mathcal{E}$, an arbitrary subset of plausible strings $\mathcal{X} = \mathcal{E} \cup \mathcal{V}$, with the other plausible strings $\mathcal{V}$ being called *valid*. We then analyze the statistical nature of these errors, and

*Email: adam@kal.ai
†Supported in part by NSF award CCF-2106444 and a Simons Investigator award. Email: vempala@gatech.edu
[1]The language model was DeepSeek-V3 (600 B parameters), accessed via the DeepSeek app on 11 May 2025.

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|---|---|
| ChatGPT: (GPT-4o) | Adam Tauman Kalai's Ph.D. dissertation (completed in 2002 at CMU) is entitled: "Boosting, Online Algorithms, and Other Topics in Machine Learning." |
| DeepSeek: | "Algebraic Methods in Interactive Machine Learning"... at Harvard University in 2005. |
| Llama: | "Efficient Algorithms for Learning and Playing Games"... in 2007 at MIT. |

Table 1: Excerpts from responses to "What was the title of Adam Kalai's dissertation?" from three popular language models.[3] None generated the correct title or year (Kalai, 2001).

apply the results for the type of errors of interest: plausible falsehoods called hallucinations. Our formalism also includes the notion of a *prompt* to which a language model must respond.

The distribution of language is initially learned from a corpus of training examples, which inevitably contains errors and half-truths. However, we show that even if the training data were error-free, the objectives optimized during language model training would lead to errors being generated. With realistic training data containing shades of error, one may expect *even higher error rates*. Thus, our lower bounds on errors apply to more realistic settings, as in traditional computational learning theory (Kearns and Vazirani, 1994).

Our error analysis is general yet has specific implications for hallucination. It applies broadly, including to reasoning and search-and-retrieval language models, and the analysis does not rely on properties of next-word prediction or Transformer-based neural networks. It only considers the two stages of the modern training paradigm: pretraining and post-training, described below. For hallucinations, taxonomies (Maynez et al., 2020; Ji et al., 2023) often further distinguish *intrinsic* hallucinations that contradict the user's prompt, such as:

> How many Ds are in DEEPSEEK? If you know, just say the number with no commentary.

DeepSeek-V3 returned "2" or "3" in ten independent trials; Meta AI and Claude 3.7 Sonnet[2] performed similarly, including answers as large as "6" and "7". Our theory also sheds light on *extrinsic* hallucinations, which contradict the training data or external reality.

### 1.1 Errors caused by pretraining

During pretraining, a *base model* learns the distribution of language in a large text corpus. We show that, even with error-free training data, the statistical objective minimized during pretraining would lead to a language model that generates errors. Proving this is nontrivial because some models make no errors, such as one that always outputs "I don't know" (IDK) or one that simply memorizes and reproduces an error-free corpus. Our analysis explains what types of errors should be expected after pretraining.

To do this, we draw a connection to binary classification. Consider questions of the form "Is this a valid language model output?" Generating valid outputs is in some sense harder than answering these Yes/No questions, because generation implicitly requires answering "Is this valid" about each candidate response. Formally, we consider the Is-It-Valid (IIV) binary classification problem which has a training set consisting of a large number of responses, each labeled either as valid (+) or error

[2] `https://meta.ai` and `https://claude.ai`, accessed May 9, 2025.

[3] The models were freely accessed 2025-05-09 via `chatgpt.com`, the DeepSeek app (R1, DeepSeek-AI et al., 2025), and `huggingface.co` (Llama-4-Scout-17B-16E-Instruct), respectively. None of the models searched the Web.

| **Valid examples** + | **Error examples** − |
|---|---|
| Greetings. | Greatings. |
| How can I help? | How kan eye help? |
| There are 2 D's in LADDER. | There are 3 L's in SPELL. |
| There is 1 N in PIANO. | There is 1 G in CAT. |
| Mia Holdner's birthday is 4/1. | Colin Merivale's birthday is 8/29. |
| I don't know Zdan's birthday. | Jago Pere's birthday is 8/21. |

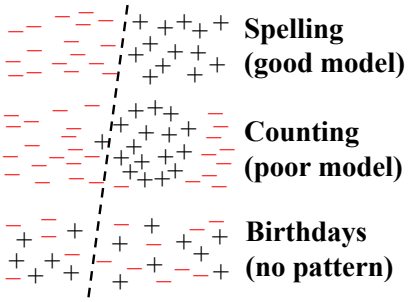


Figure 1: Is-It-Valid requires learning to identify valid generations using labeled $\pm$ examples (left). Classifiers (dashed lines) may be accurate on certain concepts like spelling (top) but errors often arise due to poor models (middle) or arbitrary facts when there is no pattern in the data (bottom).

$(-)$, as illustrated in Fig. 1. For this supervised learning problem, both train and test data are 50/50 mixtures of valid examples labeled as $+$ (i.e., the pretraining data since we assume it is valid) and uniformly random errors from $\mathcal{E}$ labeled as $-$. We then show how any language model can be used as an IIV classifier. This in turn allows us to establish a mathematical relationship between generative errors (such as hallucinations) and IIV misclassification rate:

$$\text{(generative error rate)} \gtrsim 2 \cdot \text{(IIV misclassification rate)}.$$

Language models avoid many types of errors such as spelling mistakes, and not all errors are hallucinations. The reduction from IIV misclassification to generation illuminates the statistical nature of generative errors. The analysis shows how pretraining directly contributes to errors. Furthermore, it shows that the same *statistical factors* contributing to errors in binary classification also cause language model errors. Decades of research has shed light on the multifaceted nature of misclassification errors (Domingos, 2012). Fig. 1 (right) illustrates these factors visually: top, separable data classified accurately; middle, a poor model of a linear separator for a circular region; and bottom, no succinct pattern. Section 3.3 analyzes several factors, including the following stylized setting with epistemic uncertainty, when there is no pattern in the data.

This reduction ties together earlier work which covered different types of facts. For example, Kalai and Vempala (2024) considered a special case of arbitrary facts where there is no learnable pattern in the data, like the earlier birthday hallucination example. We show how the IIV reduction covers this case and recovers their bound that the hallucination rate, after pretraining, should be at least the fraction of training facts that appear once. For instance, if 20% of birthday facts appear exactly once in the pretraining data, then one expects base models to hallucinate on at least 20% of birthday facts. In fact, our analysis strengthens their result to include prompts and IDK responses, both essential components of hallucination.

### 1.2 Why hallucinations survive post-training

The second stage, *post-training*, refines the base model, often with a goal of reducing hallucinations. While the analysis of pretraining covered errors more generally, our analysis of post-training focuses on why overconfident hallucinations are generated rather than omitting information or expressing uncertainty such as IDK. We offer a socio-technical explanation for the persistence of hallucinations after post-training and discuss how the field can suppress them.